Gauge crossover mechanisms: spectral scope, comparison tools and source evidence

A physical continuum gap requires a constructed nontrivial theory and a positive finite physical spectral edge along a controlled weak-coupling trajectory. It does **not** require a gap at every bare coupling of a chosen lattice action. The latter all-coupling property is a separate possible strengthening. Numerical bulk transitions for a Wilson action do not determine the gapped set of the distinct Kogut–Susskind operator.

The four mechanisms below preserve their source evidence and missing estimates: curvature, a centre-stabilized deformation, a three-dimensional comparison and spectral valley lifting. Correlation inequalities, finite certification and flow give possible additional tools, not an exhaustive classification of all constructive methods. Every auxiliary relaxation rate, loop/flux comparison and physical Hamiltonian gap must name its operator and sector. The conditional transfer theorem and spectral obligations own those hypotheses.

The sources motivate an expected gap across compact simple nonabelian groups; they prove no group-universality theorem here. Existing reading labels are retained, and new research is suspended during consolidation.

Evidence across groups and lattice actions

- $SU(N)$**.** Glueball masses and string tensions computed for N from 2 to about 12 are finite and extrapolate smoothly in $1/N^2$ (Lucini and Panero, Phys. Rep. **526**, 93, 2013, metadata; Lucini and Teper, JHEP **2001**(06), 050, metadata).
- G_2**.** The exceptional group has a trivial centre and still shows confinement behaviour and a first-order deconfinement transition (Pepe and Wiese, Nucl. Phys. B **768**, 21, 2007, metadata), so the centre is not what the gap rests on.
- $U(1)$**.** Four-dimensional compact $U(1)$ has a massless Coulomb phase at weak coupling (Guth, PRD **21**, 2291, 1980; Fröhlich and Spencer, CMP **83**, 411, 1982; metadata), which is why the abelian note separates the groups by the running, $b_0 = 0$ against $b_0 = 11N/(48\pi^2)$. In three dimensions compact $U(1)$ is gapped at every coupling (Polyakov, Nucl. Phys. B **120**, 429, 1977; proved by Göpfert and Mack, CMP **82**, 545, 1982; metadata), with a gap that vanishes in physical units in the continuum.

Bulk transitions depend on the action. With the standard Wilson plaquette action, "the bulk transition is a rapid cross-over if $N \leq 4$ but becomes an increasingly strong first order transition for $N \geq 5$"; for $N = 3$ a negative adjoint term avoids the bulk phase altogether (Rindlisbacher, Rummukainen and Salami, arXiv:2306.14319, passage). In the plane of fundamental and adjoint couplings, a line of first-order transitions ends at a critical point near the Wilson axis (Bhanot and Creutz, PRD **24**, 3212, 1981, metadata). A first-order transition can involve finite-volume vacuum tunnelling splittings, but the cited Wilson-action evidence alone does not prove $\delta_\infty(g_b) = 0$ for the Kogut–Susskind operator; see the corrected coupling-set analysis §4b. Such intermediate-coupling behaviour does not by itself rule out a continuum gap along $g \to 0$. The statement the conjecture needs is $\mathcal{G} \supseteq (0, g_1)$ for some $g_1 > 0$, joined to T3 along the scaling curve.

Why $N \geq 5$ differs, and what sits beside the $SU(3)$ Wilson axis

Give the lattice action a fundamental coupling β_F and an adjoint coupling β_A. In the (β_F, β_A) plane a line of first-order bulk transitions ends at a critical point (Bhanot and Creutz 1981). For small N the Wilson axis $\beta_A = 0$ passes beside the endpoint and shows a rapid crossover; from $N = 5$ it crosses the line, and the transition strengthens with N. These cited models motivate a large-N nonanalyticity; they establish no necessity theorem for every lattice action. In two

dimensions, where the lattice theory is exactly solvable, it is analytic at every finite N and has a third-order transition at a critical 't Hooft coupling at $N = \infty$ (Gross and Witten, PRD **21**, 446, 1980; Wadia, PLB **93**, 403, 1980; metadata). In four dimensions the large-N transition is first order, and at finite N it survives as a genuine transition from $N = 5$ and is smoothed into a crossover below.

For $SU(3)$ the endpoint matters twice.

- **It realizes the second-order branch.** Approaching the endpoint, "the mass of the 0^{++} glueball decreases ... while the other glueball masses appear unchanged", which is "consistent with the notion that the bulk transition line ends in a critical endpoint with the continuum limit there being a ϕ^4 theory with a diverging correlation length only in the 0^{++} channel" (Heller, PLB **362**, 123, 1995; hep-lat/9508009, abstract). This is numerical evidence for a scalar critical-endpoint interpretation, not a constructed second continuum limit or a proof of uniqueness on families avoiding that point.
- **It sits beside the point where the map places H2.** The bands note scaled the continuum ratio $r_0 m_{0^{++}} = 4.21$ to $\beta_W = 5.7$ and found $\xi/a = 0.70$. Near there the Wilson axis passes beside the endpoint, where the scalar glueball softens, so the lattice ξ/a at $\beta_W \simeq 5.7$ is likely larger, and the measured lattice value should replace the scaled one. The conditional theorem allows the remedy: blocking generates adjoint and higher-representation terms anyway, so the reference interaction Φ_0 of H2 can be taken at a small negative adjoint coupling, away from the endpoint. For $N = 3$ such an action avoids the bulk phase, as Rindlisbacher, Rummukainen and Salami report from earlier work (arXiv:2306.14319, passage).

What the problem statement itself points to

Jaffe and Witten's official description, held as a local PDF, names in its §6.6 four routes to the gap (passage), and two of them are openings 1 and 4 below:

- a "duality transformation", with Debye screening in the Coulomb gas, proved through the sine-Gordon representation, as the model;
- the quartic potential $(A \wedge A)^2$ (Feynman, Nucl. Phys. B **188**, 479, 1981), which "may be tied to curvature in the space of connections" (Singer, Phys. Scripta **24**, 817, 1981): opening 1;
- "certain quantum mechanics problems with potentials involving flat directions ... do lead to bound states" (Simon 1983): opening 4;
- the $1/N$ expansion ('t Hooft 1974): gauge theory with $SU(N)$, $SO(N)$ or $Sp(N)$ "may be equivalent to a string theory with $1/N$ as the string coupling constant", which "might give a clear-cut explanation of the mass gap and confinement and perhaps a good starting point for a rigorous proof (for sufficiently large N)". Maldacena's paper is cited there as "surprising progress along these lines for certain strongly coupled four-dimensional gauge systems with matter, but as of yet there is no effective approach to the gauge theory without fermions." Earlier, §6.1 allows that "there might some day be an asymptotic solution in a large N limit."

Chronology. Maldacena's conjecture was posted in November 1997 (hep-th/9711200); Witten's holographic account of non-supersymmetric Yang–Mills, in which "the spontaneous breaking of the center of the gauge group, magnetic confinement, and the mass gap are coded in classical geometry", in March 1998 (hep-th/9803131, abstract); supergravity glueball masses in agreement with lattice ratios in June 1998 (Csáki, Ooguri, Oz and Terning, hep-th/9806021, abstract); the Clay problems in May 2000. The statement was written with holography in hand, by one of its authors, and records no effective approach to the pure gauge theory. The usual reasons, not checked here against a primary text, are that the tractable supergravity regime is

the opposite limit from the continuum one, with Kaluza–Klein modes at the glueball scale, and that it holds at large N only.

The large-N thread. It runs through Witten's own work: the $1/N$ expansion for baryons (Nucl. Phys. B **160**, 57, 1979) and for the $U(1)$ problem (Nucl. Phys. B **156**, 269, 1979), and the large-N lattice transition with Gross (1980), the non-analyticity behind the $N \geq 5$ bulk transition above. This programme records the route and does not adopt it: the conjecture is posed for every compact simple group, so a proof for sufficiently large N would leave $SU(3)$ open unless it came with explicit control down to $N = 3$; and at large N a lattice route meets the bulk transition, which it would have to avoid with a bulk-preventing action or with the centre-stabilized volume independence of opening 2, itself a large-N construction.

Physical action, rank and limiting order

The convention is

$$S = -\frac{\hbar}{4g^2} \int F^a_{\mu\nu} F^{a\mu\nu} d^3x\, c\, dt, \qquad g^2 = \hbar g_{\mathrm{cl}}^2, \quad \lambda = g^2 N.$$

The usual large-N limit holds physical $\hbar$ and λ fixed, therefore changes g_{cl}^2. Its vacuum ribbon expansion has topological weights $N^{2-2\mathfrak{g}}$; 't Hooft, *Nucl. Phys. B* **72** (1974) 461, DOI, and Yaffe, *RMP* **54** (1982) 407, DOI, are retained at metadata level. Large-N factorization can define classical collective dynamics. Equating this with the physical $\hbar \to 0$ limit of a fixed-rank theory would change what is held fixed. Formally keeping g_{cl} fixed and $\hbar N$ fixed is another parameter path, not that usual limit.

Conditional on a gap proportional to the RG scale, the weak-coupling running gives the schematic energy

$$\Delta \;\propto\; \frac{\hbar c}{a} \exp\left[-\frac{24\pi^2}{11\hbar g_{\mathrm{cl}}^2 N}\right]$$

times a power of the dimensionless coupling. This is scaling orientation, not a constructed spectrum or a fixed-volume asymptotic theorem. Its explicit $\hbar c/a$ prefactor survives outside the function of λ; the full energy therefore does not depend only on $\hbar N$. In the separately controlled zero-mode truncation, $\delta_1 g^{2/3} \hbar c/L$ instead equals $\delta_1 \hbar^{4/3} g_{\mathrm{cl}}^{2/3} c/L$. Matching that truncation to the full small-volume field theory requires the renormalized transfer estimates.

Four scoped mechanisms

Opening 1: curvature

The configuration space of lattice gauge theory is a product of copies of the group, and a compact simple group carries positive Ricci curvature: for $SU(N)$ with the metric $\langle X, Y\rangle = \mathrm{Tr}(XY^*)$, $\mathrm{Ric} = \frac{N}{2}\, g$. The Bakry–Émery criterion turns a positive lower bound on $\mathrm{Ric} + \mathrm{Hess}\, S$ into a log-Sobolev and a Poincaré inequality, that is into a spectral gap of the Langevin dynamics and exponential decay of correlations (Bakry and Émery, LNM 1123, 177, 1985, metadata). A flat group, $U(1)$, gets nothing from the first term, which lines up with the abelian failure.

Shen, Zhu and Zhu use exactly this to prove, for lattice Yang–Mills with 't Hooft coupling βN in any dimension $d > 1$, uniqueness of the infinite-volume measure, log-Sobolev and Poincaré inequalities and exponential decay of correlations, "a strictly positive mass gap", for $|\beta| < 1/(16(d-1))$ when the group is $SU(N)$ (CMP **400**, 2022; arXiv:2204.12737, abstract). In the Wilson normalization of this repository, $\beta_W = 2N/g^2$ with weight $e^{(\beta_W/N)\sum \mathrm{Re}\,\mathrm{Tr}\, U_p}$, their β is β_W/N^2, so for $SU(3)$ in four dimensions the condition is $\beta_W < 3/16$, that is

$$g^2 > 32.$$

That conversion is ours and must be checked against their definitions. If it stands, the curvature method already beats this repository's strong-coupling sufficient Wilson thresholds, $g^2 \geq 1059$ for the all-representation polymer bound (note) and $g^2 > 444$ by Dobrushin (note). The former Wilson figure 176 was a leading-activity estimate; the former Kogut–Susskind figure 388 was withdrawn on October 2 (note). The latter keeps an existential volume-uniform strong-coupling theorem and an explicit finite-volume bound, not a certified numerical uniform threshold. If the curvature conversion and physical transfer stand, the comparison narrows band B of the bands note from $\beta_W \in (0.0135, 5.7)$ to $\beta_W \in (0.19, 5.7)$.

The weak side is where the multiscale form enters. Bauerschmidt and Bodineau derive "a multiscale generalisation of the Bakry–Émery criterion" through the Polchinski renormalization equation, "effective for measures which are far from log-concave", and prove optimal log-Sobolev inequalities for the continuum sine-Gordon model at $\beta < 6\pi$ (CPAM **74**, 2020; arXiv:1907.12308, abstract). Its output is a spectral gap controlled at every scale, which is the form of hypothesis H1 in the conditional theorem. It also sits outside the family qualified in flow-conjugation: that note shows invariance when the same decomposition is conjugated; a new splitting or a curvature bound needs a separate analysis.

The risk is concrete. After a blocking step the effective action is a function on the coarse configuration manifold, and the criterion needs $\mathrm{Ric} + \mathrm{Hess}\, S_{\mathrm{eff}}$ bounded below at every scale. At weak coupling the Hessian is degenerate along gauge orbits and along the abelian valleys, and the curvature of the group has to carry those directions. The conversion to $g^2 > 32$, the precise operator controlled and the curvature budget against the Wilson Hessian remain unchecked in this treatment. Multiscale transport requires a named lower bound preserved by the actual generated interaction.

Opening 2: the centre-stabilized theory at small radius

Ünsal and Yaffe add a double-trace deformation that keeps centre symmetry unbroken; on $\mathbb{R}^3 \times S^1$ at small radius "the deformed Yang–Mills theory has a mass gap and exhibits linear confinement", by an analytic semiclassical treatment, and "there are no order parameters which distinguish the small and large radius regimes" (PRD **78**, 065035, 2008, abstract). The mechanism at small radius is Polyakov's monopole plasma, which Göpfert and Mack made rigorous for three-dimensional compact $U(1)$ at every coupling. A rigorous estimate with explicit constants would concern that deformed theory. Carrying it to undeformed $\mathbb{R}^4$ Yang–Mills remains an additional adiabatic-continuity conjecture.

Opening 3: three dimensions first

In the convention of the three-dimensional treatment, $\lambda_3 = \hbar g_{\mathrm{cl}}^2$ has inverse-length units, $E_3 = \hbar c \lambda_3$ is the energy scale and the dimensionless lattice coupling is $\lambda_3 a$. Superrenormalizable ultraviolet scaling supplies a different problem from four-dimensional logarithmic running. The corrected Gaussian remainder diagnostic proves no three-dimensional closure or numerical threshold by dimensional substitution. A physical positive lower bound on the scaling function, nonzero-mode control and a continuum construction remain distinct open requirements.

Opening 4: the Planck-gap link

The small-volume mechanism of G07 and G08 is Simon's: along the flat valleys of the commutator potential the transverse modes are oscillators whose frequency grows with the distance along the valley, and their zero-point energy lifts the valley (Simon, Ann. Phys. **146**, 209, 1983, metadata). In its simplest form, for fixed x,

$$\frac{\hat{p}_y^2}{2m} + \frac{1}{2} m \Omega(x)^2 \hat{y}^2 \;\geq\; \frac{1}{2} \hbar\, \Omega(x),$$

an operator inequality that holds in every state. The torus-valley note finds the field-theory version term by term: the linear term $2|a|$ of the valley potential is the zero-point energy of the $k = 0$ charged modes.

The Planck-gap work sharpens what kind of floor this is. Its floors that rest on second moments of a prepared state fall to squeezing and to grid states, and the ones that survive rest on the commutator alone: the phase between Newton's polygon and the parabola, and the cost of disturbance. Simon's inequality is of the surviving kind, a spectral statement, so the valley lifting is the mass-gap instance of the lesson the Planck paper draws. The missing target is hypothesis (H3) of the Feshbach note, the gap of the zero-mode Hamiltonian h_3 plus the even valley potential $U(a)$, proved by a Simon-type operator inequality with every constant explicit. That is a conditional lower-side target in the small-volume regime; its nonzero-mode transport is not proved by the oscillator inequality alone.

Correlation inequalities and the transfer they require

If Wilson loop expectations increase with β_W, an area-law string tension defined by the corresponding limit is nonincreasing where the limit and comparison are valid. A lower area-law bound can then transfer from a weaker to a stronger coupling. Monotonicity of a loop expectation alone does not imply monotonicity of a connected correlator, its decay exponent or the vacuum-sector spectral gap. Both terms of a truncated correlator may increase. The old claim that this closes every stronger-coupling channel is withdrawn.

Ginibre's method needs the interaction to lie in a cone of products of single-site functions satisfying the duplicate-variables condition. For the character cone of $SU(2)$ the single-site condition reduces to

$$\iint \prod_{i=1}^{n} \left(\chi_i(x) - \chi_i(y)\right) dx\, dy = \sum_{S \subseteq \{1..n\}} (-1)^{|S^c|}\, N_S\, N_{S^c}, \qquad N_S = \dim \mathrm{Hom}\left(1, \bigotimes_{i \in S} V_i\right),$$

which vanishes for odd n, equals $2N_{1234} + 2(\delta_{12}\delta_{34} + \delta_{13}\delta_{24} + \delta_{14}\delta_{23}) \geq 0$ for $n = 4$, and is positive in every case computed at $n = 6$ (six spin $\frac{1}{2}$: 70; six spin 1: 100; four spin $\frac{1}{2}$ and two spin 1: 28; six spin 2: 260). The single-site condition is therefore no obstruction as far as checked. The obstruction is that the plaquette term $\mathrm{Re}\,\mathrm{tr}(U_1U_2U_3U_4)$ is not a nonnegative combination of products of single-link functions: the abelian proof writes $\cos(\theta_1 + \dots)$ through sums and differences of angles, which uses the commutativity of the group, and no usable positive plaquette decomposition for the matrix product is established by the examined construction. The second Griffiths inequality was not established by the sources read here for $SU(2)$ and $SU(3)$ lattice gauge theory; no proof or counterexample is supplied here, and the centre-based inequalities that do exist are listed in the tools-and-source table below.

The finite checks are not an all-order character-cone theorem. The obstacle identified here is the absence of the needed positive plaquette decomposition in the examined construction, not a proof that every correlation inequality for a nonabelian theory is impossible.

Tools and source scope

These are the sources previously cited in this family, at **metadata level** unless a linked canonical proof records a deeper read. They are leads and scope comparisons, not independently verified universal impossibility claims.

Tool/source	Potential input	Missing transfer
Ginibre, CMP 16 (1970) 310; Aizenman–Barsky–Fernández, JSP 47 (1987) 343	Positive cones and Ising-type sharpness	A nonabelian plaquette interaction satisfying the required hypotheses
Mack–Petkova, Ann. Phys. 123 (1979) 442 and 125 (1980) 117; Tomboulis–Yaffe, CMP 100 (1985) 313	Centre, electric-flux and vortex/loop comparisons	A vacuum-sector physical spectral bound; a string tension is a different observable
Fröhlich–Israel–Lieb–Simon, CMP 62 (1978) 1	Reflection-positive chessboard bounds for specified events	Good-region mixing and a normalized block free-energy estimate
Fröhlich–Simon–Spencer, CMP 50 (1976) 79	Gaussian-domination/infrared techniques for their stated spin interactions	The needed domination for the nonquadratic plaquette product; nonabelian target curvature alone is not the failure reason
Knabe, JSP 52 (1988) 627	A finite-size gap criterion under frustration-free hypotheses	Those hypotheses are not verified for interacting KS at general coupling; this does not exhaust all finite-size criteria
Finite mixing proposal, Dobrushin proof	A certified supremum over boundary conditions could imply a Wilson-transfer gap	No certificate at the physical crossover or for the generated interaction is supplied
Exact flow conjugation	Spectrum-preserving change of variables	New effective decomposition, relative error and iterative control
Small-volume bridge	Compressed zero-mode estimates with a named complement/Schur target	Renormalized cutoff-uniform transfer and passage to large physical volume

A signed spin-foam representation does not make the original Wilson probability weight negative, and positive plaquette coefficients do not make every recoupling amplitude positive. Centre comparisons, electric-flux sectors and glueball correlations must name the sector they control. The 2007 approximate centre-decimation proposal (arXiv:0707.2179, metadata) was not established as a theorem by this repository's reading.

Gauge fixing can change a conditional specification; merely subtracting a maximal-tree fraction from a neighbour count does not prove a new Dobrushin coefficient. The earlier numerical suggestion $444 \mapsto 330$ is unsupported here. Likewise, neither a perturbative four-dimensional Langevin construction nor its obstruction is proved by this source survey; an auxiliary stochastic clock is not a physical energy gap.

Finite certification and the actual interaction

The earlier illustrative boxes at $\beta_W \simeq 6$ had roughly 216–2000 links, hence thousands of integration variables. These are scale estimates from the bands note, not a complexity theorem excluding every certification method. Monte Carlo by itself is not a certified supremum over boundary conditions. A rigorous polymer enumeration needs an analytic tail and its own convergence region. The explicit Wilson criteria here prove sufficient strong-coupling regions, not limits on all possible future methods. The strong-coupling proofs and Dobrushin criterion own their constants.

The weak side has exact Gaussian and conditional remainder tests in Gaussian blocking. The corrected covariance sum and soft/hard distinction supply no certified nonlinear threshold. The mid-plane notebook later proves different retained-barrier weak-strip estimates. Their coefficients and cutoff/volume hypotheses cannot be silently substituted for a physical-coupling full integral.

After blocking, all generated interactions must be controlled. There is no proved arrival in the completely analytic set at order-one coupling and no proved universal statement that all difficulty is one fixed unknown lattice interaction. A possible mixing certificate at that scale would still need stable transport from the weak trajectory, physical spectral visibility, nontriviality and continuum reconstruction.

Consequence for STATE

The examined mechanisms retain a concrete boundary: curvature normalization and physical transfer, removal of the deformation, continuum/nonzero-mode control, or a normalized state-uniform operator estimate. Loop monotonicity, signed recoupling, finite certification proposals and unitary flow do not supply those steps automatically. No exhaustive impossibility result or certified intermediate-coupling arrival is asserted. Physical action, rank, volume, lattice cutoff and auxiliary time retain distinct limiting orders.